

e^x and e^{-x} appear a lot in math, so there are some nice notations for it.

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

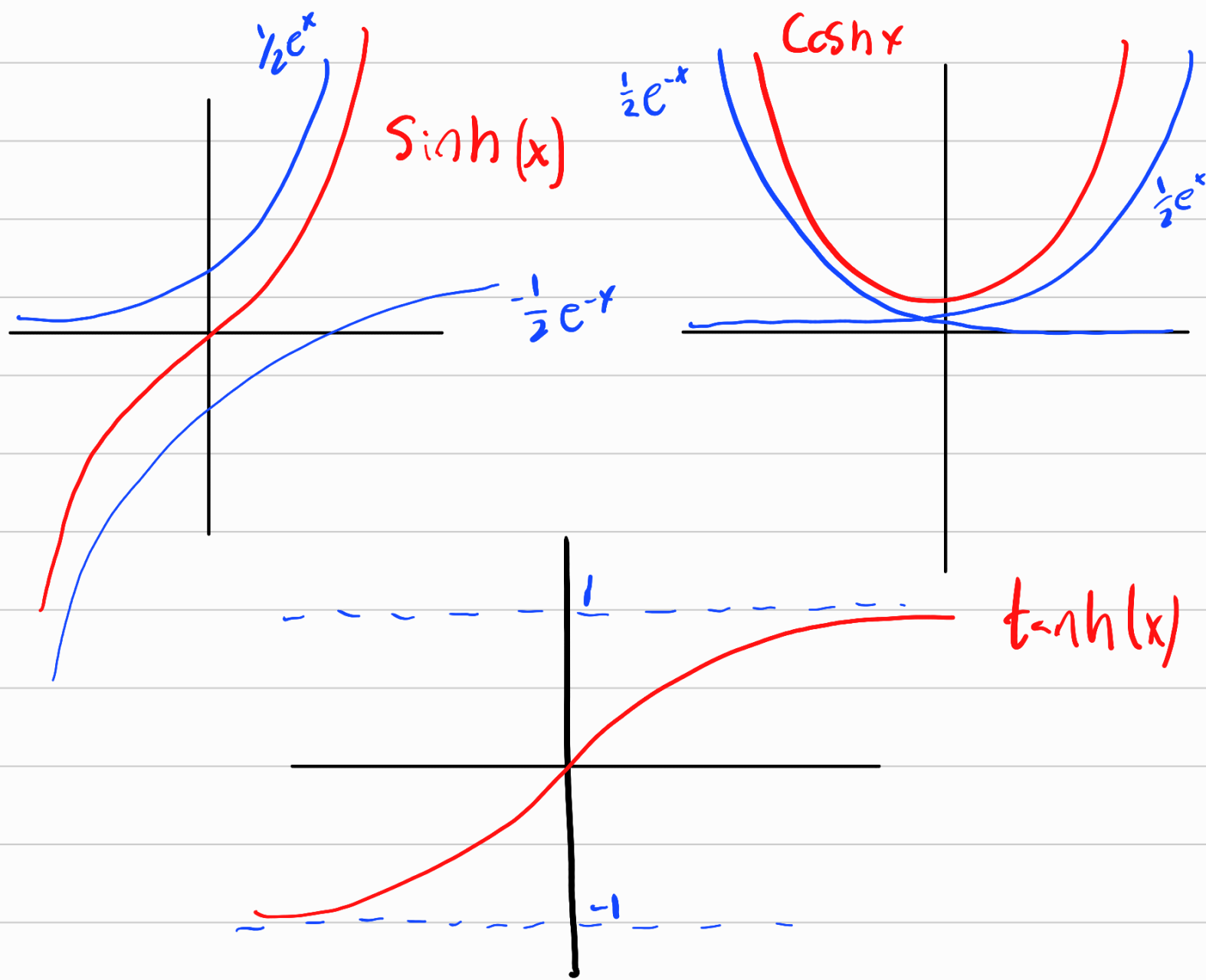
$$\operatorname{csch} x = \frac{1}{\sinh x}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\operatorname{coth} x = \frac{\cosh x}{\sinh x}$$



Classic Identities

$$\cosh^2 x - \sinh^2 x = 1$$

$$1 - \tanh^2 x = \operatorname{sech}^2 x$$

Pf/

Immediate by definitions.

derivatives

$$\frac{d}{dx} \sinh x = \frac{d}{dx} \frac{e^x - e^{-x}}{2}$$

$$= \frac{e^x + e^{-x}}{2}$$

$$= \cosh x$$

In fact, all derivatives are almost analogous to $\sin, \cos, \tan$, etc

$$\frac{d}{dx} \sinh(x) = \cosh(x), \quad \frac{d}{dx} \cosh(x) = \sinh(x) \quad \frac{d}{dx} \tanh(x) = \operatorname{sech}^2(x)$$

$$\frac{d}{dx} \cosh x = \sinh(x), \quad \frac{d}{dx} \operatorname{sech}(x) = -\operatorname{sech}(x) \tanh(x)$$

$$\frac{d}{dx} \tanh(x) = \operatorname{sech}^2(x), \quad \frac{d}{dx} \coth(x) = -\operatorname{csch}^2(x)$$

And for inverse hyperbolic:

$$\frac{d}{dx} \operatorname{arcsinh}(x) = \frac{1}{\sqrt{1+x^2}}, \quad \frac{d}{dx} \operatorname{arccsch}(x) = \frac{-1}{|x|\sqrt{x^2+1}}$$

$$\frac{d}{dx} \operatorname{arcosh}(x) = \frac{1}{\sqrt{x^2-1}}, \quad \frac{d}{dx} \operatorname{arsech}(x) = \frac{-1}{x\sqrt{1-x^2}}$$

$$\frac{d}{dx} \operatorname{artanh}(x) = \frac{1}{1-x^2} \quad |x| < 1, \quad \frac{d}{dx} \operatorname{arcoth}(x) = \frac{1}{1-x^2} \quad |x| > 1$$

Note: $\operatorname{arcsinh}(x) = \ln(x + \sqrt{x^2+1}) \quad x \in \mathbb{R}$

$$\operatorname{arcosh}(x) = \ln(x + \sqrt{x^2-1}) \quad x \geq 1$$

$$\operatorname{artanh}(x) = \frac{1}{2} \ln\left(\frac{1+x}{1-x}\right) \quad -1 < x < 1$$

Let $y = \operatorname{arcsinh}(x)$, so $x = \sinh y = \frac{e^y - e^{-y}}{2}$

$$e^y - 2x - e^{-y} = 0$$

$$e^{2y} - 2xe^y - 1 = 0$$

$$u = e^y, \quad u^2 - 2xu - 1 = 0$$

$$e^y = u = \frac{2x \pm \sqrt{4x^2 + 4}}{2} = x \pm \sqrt{x^2 + 1}$$

$$e^y \geq 0, \text{ so } e^y = x + \sqrt{x^2 + 1}$$

$$y = \ln(x + \sqrt{x^2 + 1})$$

pf/ $\frac{d}{dx} (\operatorname{arcsinh}(x))$

$$y = \operatorname{arcsinh}(x), \quad \sinh(y) = x$$

↙ implicit derivative

$$y' \cosh(y) = 1$$

$$y' = \frac{1}{\cosh(y)} = \frac{1}{\sqrt{1 - \sinh^2(y)}} = \frac{1}{\sqrt{1 - x^2}}$$